Exercise 3.7

1. Find the square root of the following rational expressions.

(i) $\dfrac{400x^4y^{12}z^{16}}{100x^8y^4z^4}$ (ii) $\dfrac{7x^2+2\sqrt{14}x+2}{x^2-\frac{1}{2}x+\frac{1}{16}}$ (iii) $\dfrac{121(a+b)^8(x+y)^8(b-c)^8}{81(b-c)^4(a-b)^{12}(b-c)^4}$

2. Find the square root of the following

(i) $4x^2+20x+25$ (ii) $9x^2-24xy+30xz-40yz+25z^2+16y^2$

(iii) $(4x^2-9x+2)\,(7x^2-13x-2)\,(28x^2-3x-1)$

(iv) $\left(2x^2+\dfrac{17}{6}x+1\right)\left(\dfrac{3}{2}x^2+4x+2\right)\left(\dfrac{4}{3}x^2+\dfrac{11}{3}x+2\right)$

3.5.2 Finding the Square Root of a Polynomial by Division Method

The long division method in finding the square root of a polynomial is useful when the degree of the polynomial is higher.

Example 3.21 Find the square root of $64x^4-16x^3+17x^2-2x+1$

Solution

$$
\begin{array}{r|l|l}
 & 8x^2-x+1 & \\
\hline
8x^2 & 64x^4-16x^3+17x^2-2x+1 & \\
 & 64x^4 & (-)\\
\hline
16x^2-x & \quad -16x^3+17x^2 & \\
 & \quad -16x^3\ \ +x^2 & (-)\\
\hline
16x^2-2x+1 & \qquad\qquad 16x^2-2x+1 & \\
 & \qquad\qquad 16x^2-2x+1 & (-)\\
\hline
 & \qquad\qquad\qquad 0 &
\end{array}
$$

Therefore, $\sqrt{64x^4-16x^3+17x^2-2x+1}=\left|8x^2-x+1\right|$

Note

Before proceeding to find the square root of a polynomial, one has to ensure that the degrees of the variables are in descending or ascending order.

Example 3.22 If $9x^4+12x^3+28x^2+ax+b$ is a perfect square, find the values of a and b.

Solution

$$
\begin{array}{r|l|l}
 & 3x^2+2x+4 & \\
\hline
3x^2 & 9x^4+12x^3+28x^2+ax+b & \\
 & 9x^4 & (-)\\
\hline
6x^2+2x & \quad 12x^3+28x^2 & \\
 & \quad 12x^3+4x^2 & (-)\\
\hline
6x^2+4x+4 & \qquad\qquad 24x^2+ax+b & \\
 & \qquad\qquad 24x^2+16x+16 & (-)\\
\hline
 & \qquad\qquad\qquad 0 &
\end{array}
$$

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